

Retail Sales Optimization

Data Analytics & Business Intelligence Project

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PYTHON

SQL

POWER BI

PANDAS

NUMPY

MATPLOTLIB



Made with GAMMA



Executive Summary

This project analyzes retail sales transactions to identify customer purchasing patterns, product performance, seasonal trends, and revenue opportunities. The solution includes data cleaning, feature engineering, exploratory data analysis, SQL-based business analysis, Power BI dashboard development, and business recommendations.

Data Cleaning

Missing values treated, duplicates removed, dates converted to DateTime format

Feature Engineering

Created Age Category, Month, Year, and recalculated Total Amount

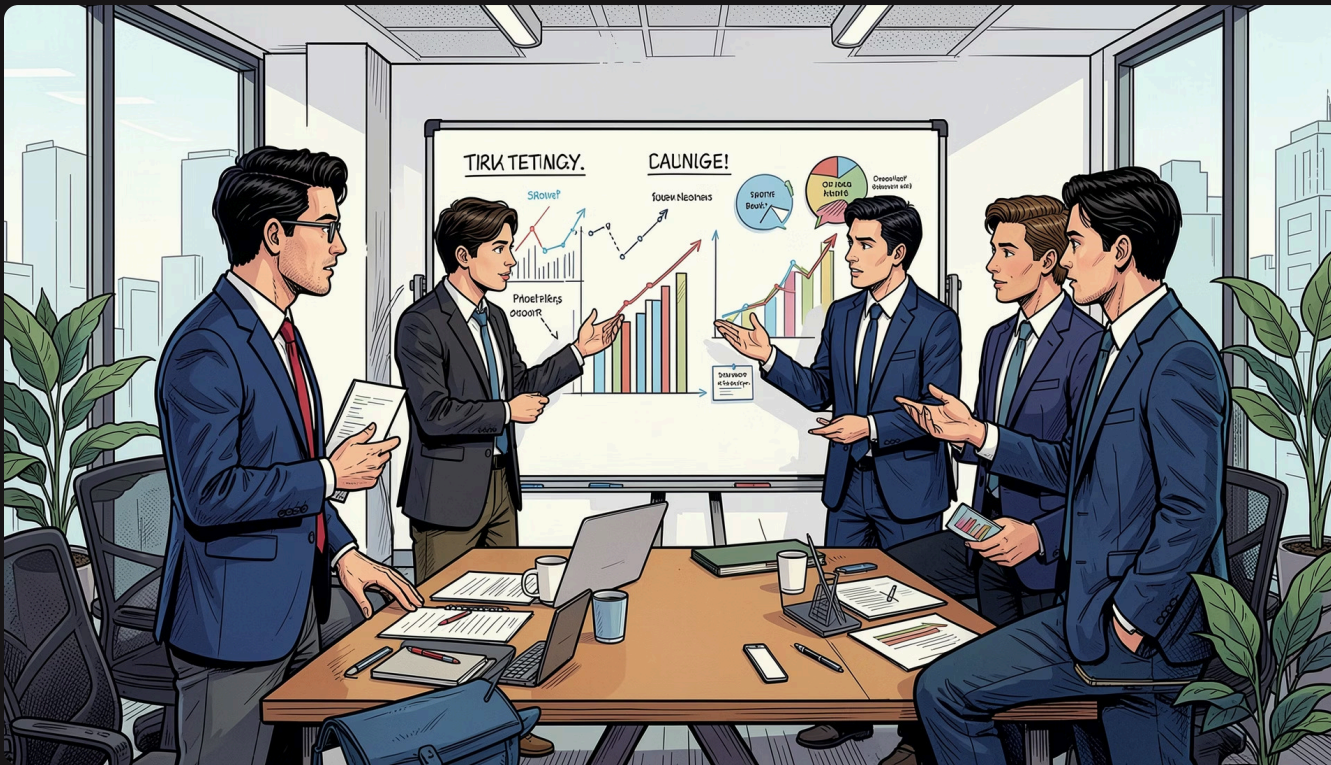
EDA & SQL Analysis

Exploratory data analysis and SQL-based business queries

Power BI Dashboard

Interactive dashboard with slicers and visualizations

Business Problem



The core objective: determine which categories, customer segments, and time periods drive the most revenue — and identify opportunities to improve sales performance.

- Highest Revenue Categories
- Best Customer Segments
- Peak Sales Months
- Age-Group Purchasing Patterns
- Opportunities to Improve Sales Performance

Dataset Overview

1,012

Transaction Records

Retail transaction records in the dataset

9

Features

Including customer demographics, product category, quantity, price per unit, and total amount

Data Cleaning

- Missing values treated
- Duplicates removed
- Dates converted to DateTime format
- Total Amount recalculated: **Quantity × Price per Unit**

Feature Engineering

- Created Age Category
- Created Month & Year fields
- Recalculated Total Amount

These engineered features improve business analysis by enabling segmentation by age group and time period.

Exploratory Data Analysis

Electronics

Generated the **highest revenue** among all product categories

Clothing

Achieved the **highest quantity sold** across all categories

Female Customers

Generated **slightly higher revenue** than male customers

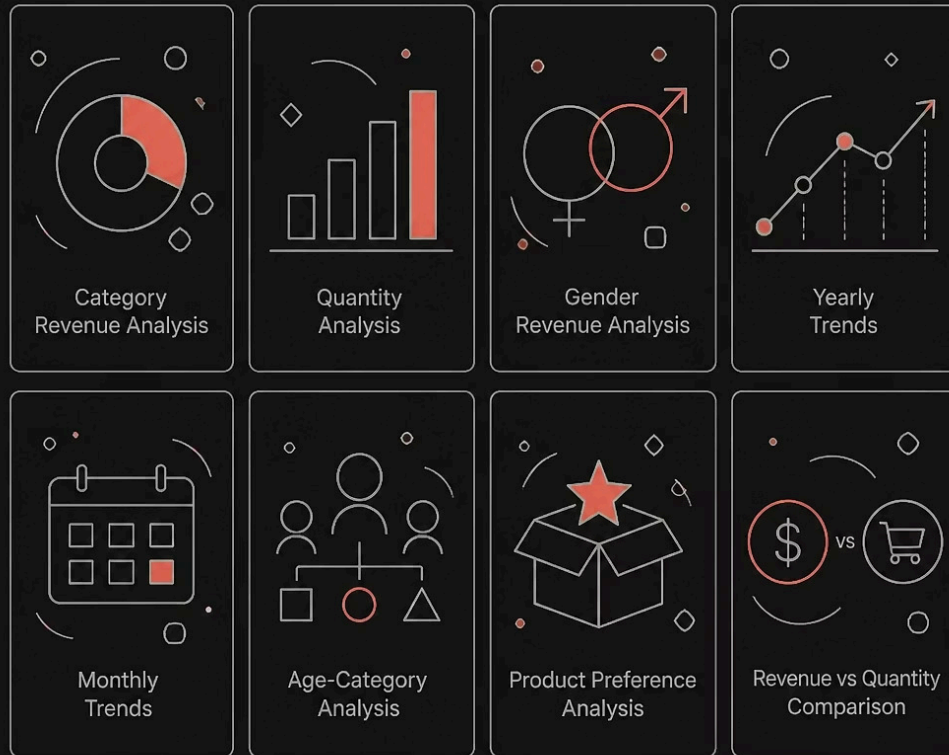
Middle-Aged Customers

Were the **most profitable** age-group segment

Peak Sales Months

May, October, and December were the top revenue months

SQL Business Analysis



What We Queried

Using MySQL and SQL, the following business analyses were performed on the retail transaction dataset:

- Category revenue analysis
- Quantity analysis
- Gender revenue analysis
- Yearly and monthly trends
- Age-category analysis
- Product preference analysis
- Revenue vs quantity comparison

These queries enabled deep exploration of purchasing patterns, customer segments, and seasonal trends.

Power BI Dashboard

An interactive dashboard was developed with slicers and visualizations to explore retail sales data dynamically.



Sales by Gender

Revenue breakdown comparing female and male customer segments



Sales by Category

Revenue and quantity distribution across product categories



Sales by Age Category

Revenue contribution segmented by customer age groups



Monthly Trends

Time-series view of sales performance across months



Key Business Insights



Electronics Leads Revenue

Electronics is the leading revenue category, generating the highest total sales



Clothing Leads Volume

Clothing leads in sales volume with the highest quantity sold



Female Customers Outspend

Female customers slightly outspend male customers in total revenue



Middle-Aged Most Profitable

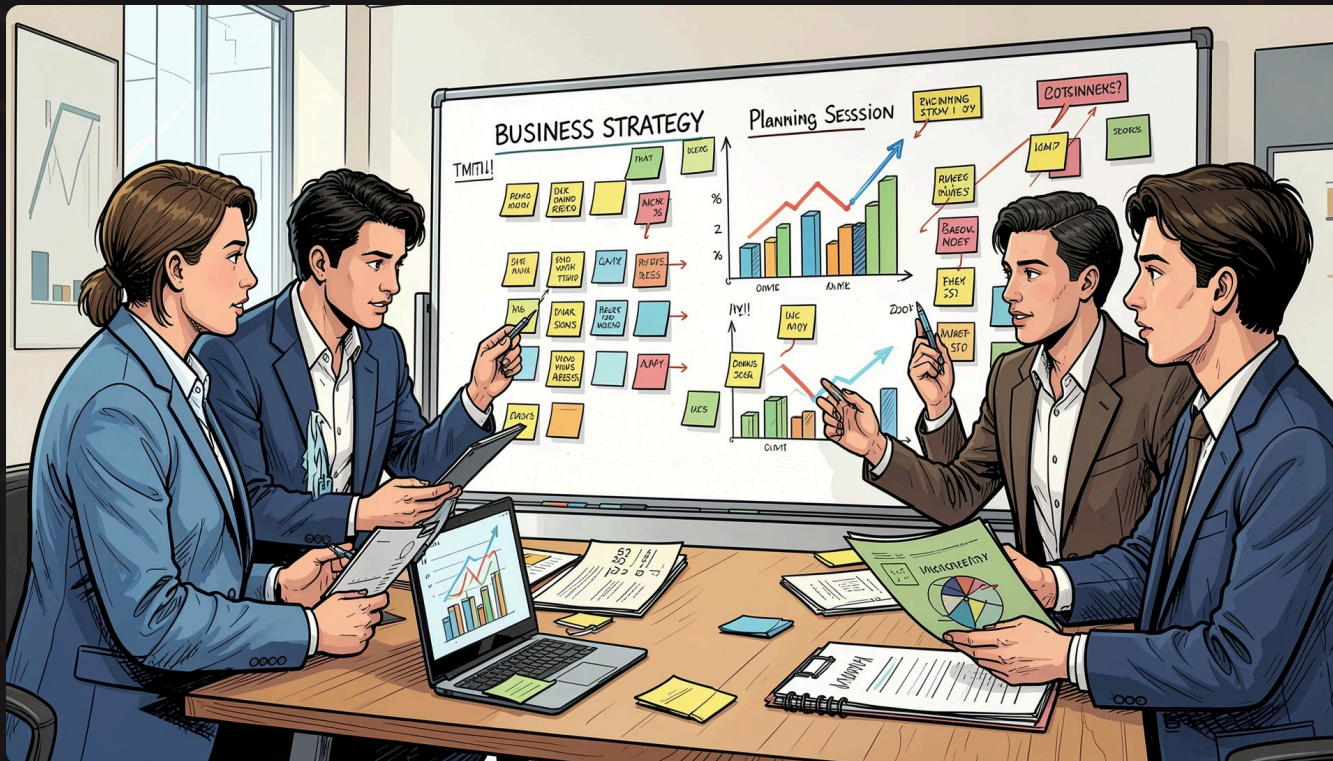
Middle-aged customers generate the most revenue across all segments



Peak Months: May, Oct, Dec

Peak sales occur in May, October, and December

Business Recommendations



1 Boost Electronics Promotions

Increase marketing spend on the highest-revenue category

2 Target Middle-Aged Customers

Focus campaigns on the most profitable age segment

3 Optimize Inventory Before Peaks

Stock up before May, October, and December

4 Improve Beauty Category Marketing

Enhance marketing efforts for the beauty category

5 Enhance Customer Segmentation

Develop more granular customer segments for targeted strategies

6 Implement Loyalty Programs

Launch loyalty programs to drive repeat purchases



Conclusion

This project transformed raw retail transaction data into actionable business intelligence using Python, SQL, and Power BI, enabling data-driven retail decision making.

Python & Pandas

Data cleaning, feature engineering, and EDA

MySQL & SQL

Business analysis and trend queries

Power BI

Interactive dashboard and visual storytelling